

JINLONG XUE

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🎓 EDUCATION

Beijing University of Posts and Telecommunications (BUPT), Beijing, China 2022 – 2025

Master student in Artificial Intelligence (AI), Advised by Prof. Ya Li

- GPA: 3.76 / 4.0 (89.52 / 100)
- English: IELTS 7.0; GRE 323+3.5
- Research on Speech Synthesis (TTS) and Audio Synthesis (TTA)
- Coursework: Computational Linguistics, Machine Vision, Speech Interaction

Beijing University of Posts and Telecommunications (BUPT), Beijing, China 2018 – 2022

B.S. in Computer Science (CS)

- GPA: 3.77 / 4.0 (90.61 / 100) ; Rank: 18/386 (4%)
- Coursework: Computer Networks, Operating System, Information and Knowledge Acquisition, Probability Theory and Mathematical Statistics, Discrete Mathematics, Machine Learning, Data Structures, Python

Massachusetts Institute of Technology (MIT), Boston, USA August. 2019

On-campus EECS Course in Machine Learning and Artificial Intelligence

- Coursework: Fundamentals of Machine Learning, Deep Learning, Computer Vision

👤 EXPERIENCE

MBZUAI Abu Dhabi, UAE August. 2024 – Present

Visiting Student Supervised by Pro. Zhiqiang Shen

Introduction: Research on the large multi-modal model

- Survey on state-of-the-art large vision-language model such as LLaVA, Cambrian-1, Mini-Gemini
- Lead the vision project and conduct the entire pretrain and finetune process for 0.5/1.5/7/8B vision-language models, including various LLaVA and LLaVA-NeXT configurations
- Evaluate visual models on multiple metrics, including the assessment of utmost 65B LLM-based models
- Explore large-scale multi-modal model with improved comprehension and advanced generative capabilities

01.AI Inc. Beijing, China May. 2024 – August. 2024

Research Intern Multi-Modal AI

Introduction: Research on the latest discrete-token-based speech synthesis

- Survey and implementation on latest large-scale speech synthesis models, such as VALL-E, SoundStorm
- Integrate TTS model into digital avatars service, allowing for natural speech cloning

Unisound Inc. Beijing, China Jan. 2022 – June. 2022

Intern College-Enterprise Cooperation

Introduction: Research on Chinese multi-emotion speech synthesis

- Study and reproduce latest papers related to multi-emotion and expressive speech synthesis
- Propose a combination of explicit and implicit emotion control TTS methods, allowing implicit control through reference audio as well as fine-grained explicit control of different emotional levels

🏆 AWARDS

Huawei Enterprise Scholarship 2023

Graduate First-class Scholarship 2022 & 2023

National College Student Mathematics Competition (Third Prize) 2020

Interdisciplinary Contest In Modeling (Honorable Mention) 2020

📖 PUBLICATIONS

2024

- **Auffusion: Leveraging the Power of Diffusion and Large Language Models for Text-to-Audio Generation**
Jinlong Xue, Yayue Deng, Yingming Gao, Ya Li
Project Page: <https://auffusion.github.io> (with more than **140** stars)
Accepted by IEEE/ACM Transactions on Audio Speech and Language Processing (TASLP), 2024
- **Improving Audio Codec-based Zero-Shot Text-to-Speech Synthesis with Multi-Modal Context and Large Language Model**
Jinlong Xue, Yayue Deng, Yichen Han, Yingming Gao, Ya Li
Accepted by Interspeech, 2024 (Oral)
- **Retrieval Augmented Generation in Prompt-based Text-to-Speech Synthesis with Context-Aware Contrastive Language-Audio Pretraining**
Jinlong Xue*, Yayue Deng*, Yingming Gao, Ya Li
Accepted by Interspeech, 2024
- **CONCSS: Contrastive-based Context Comprehension for Dialogue-appropriate Prosody in Conversational Speech Synthesis**
Yayue Deng, **Jinlong Xue**, Yukang Jia, Qifei Li, Yingming Gao, Dengfeng Ke, Ya Li, et al.
Accepted by IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2024
- **Frame-level emotional state alignment method for speech emotion recognition**
Qifei Li, Yingming Gao, Cong Wang, Yayue Deng, **Jinlong Xue**, Yichen Han, Ya Li
Accepted by IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2024

2023

- **M2-CTTS: End-to-End Multi-Scale Multi-Modal Conversational Text-to-Speech Synthesis**
Jinlong Xue, Yayue Deng, Fengping Wang, Ya Li, Yingming Gao, Jianhua Tao, et al.
Accepted by IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2023
- **CMCU-CSS: Enhancing Naturalness via Commonsense-based Multi-modal Context Understanding in Conversational Speech Synthesis**
Yayue Deng*, **Jinlong Xue***, Fengping Wang, Yingming Gao, Ya Li
Accepted by ACM International Conference on Multimedia (MM), 2023

2022

- **ECAPA-TDNN for Multi-speaker Text-to-speech Synthesis**
Jinlong Xue, Yayue Deng, Yichen Han, Ya Li, Jianqing Sun, Jiaen Liang
Accepted by IEEE International Symposium on Chinese Spoken Language Processing (ISCSLP), 2022
- **Rhythm-controllable Attention with High Robustness for Long Sentence Speech Synthesis**
Dengfeng Ke, Yayue Deng, Yukang Jia, **Jinlong Xue**, Qi Luo, Ya Li, et al.
Accepted by IEEE International Symposium on Chinese Spoken Language Processing (ISCSLP), 2022
- **A Keypoint Based Enhancement Method for Audio Driven Free View Talking Head Synthesis**
Yichen Han, Ya Li, Yingming Gao, **Jinlong Xue**, Songpo Wang, Lei Yang
Accepted by IEEE 24th International Workshop on Multimedia Signal Processing (MMSP), 2022

(* equal contribution)

⚙️ SKILLS

- **Research:** Experienced in deep learning / data analysis, proficient in Pytorch / Numpy / Matplotlib
- **Programming:** Proficient in Python, also ability with C++ / C / JavaScript / LaTeX / SQL
- **Languages:** Fluent in English, native in Chinese - Mandarin
- **Others:** Good communication skills, love collaborating with people, works well in a team